In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

FIITJEE

JEE (Advanced), 2017

ALL INDIA TEST SERIES

FULL TEST – IV

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 180**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 10)** contains 10 multiple choice questions which have only one correct answer. Each question carries **+2 marks** for correct answer. There is no negative marking.

Section-A (11 to 15) contains 5 multiple choice questions which have one or more than one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) **Section-C (01 to 05)** contains 5 Numerical based questions with answers as numerical value and each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

Straight Objective Type

This section contains **10 multiple choice questions** numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct.

1. A cylindrical block of radius R, height h and density ρ is floating in liquid of density 2ρ contained in a cylindrical container of radius 2R. Time period of small vertical oscillations of the block is

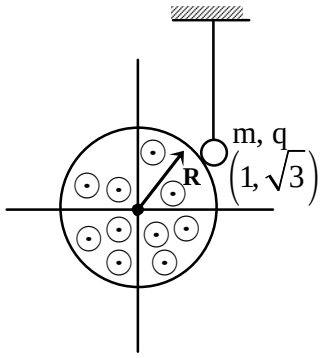
(A) $2\pi\sqrt{\frac{5h}{6g}}$ (B) $\pi\sqrt{\frac{3h}{2g}}$

(C) $2\pi\sqrt{\frac{h}{2g}}$ (D) $\pi\sqrt{\frac{h}{2g}}$

2. A charge particle of mass m and charge q is suspended with the help of a string having breaking strength of 2mg and mass having co-ordinates $\left(1, \sqrt{3}\right)$ while the space carries a uniform magnetic field in cylindrical region with centre at origin and radius R = 2 m. Maximum charge q if string does not break at the instant magnetic field start increasing with rate 0.5 Tesla per second is

(A) 4mg (B) 8mg

(C) 2mg (D) None



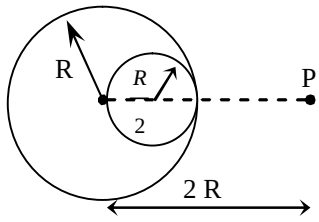
3. In a certain region of space gravitational field is given by $g = \left(-\frac{K}{r}\right)$. Taking potential at $r = r_0$ to be V_0 the potential at any general point is given by

(A) $V = K\ln\left(\frac{r}{r_0}\right) + V_0$ (B) $V = K\ln\left(\frac{r}{r_0}\right) - V_0$

(C) $V = K\ln\left(\frac{r_0}{r}\right) + V_0$ (D) $V = K\ln\left(\frac{r_0}{r}\right) - V_0$

Space for rough work

4. A solid sphere of uniform density and radius R exerts a gravitational force F_1 on a particle at P. A spherical cavity of radius $\frac{R}{2}$ is now removed from the sphere. Now the same sphere (with cavity) exerts a gravitational force F_2 on the same particle at P. Then, $\frac{F_1}{F_2}$ is



(A) $\frac{9}{7}$ (B) $\frac{7}{9}$

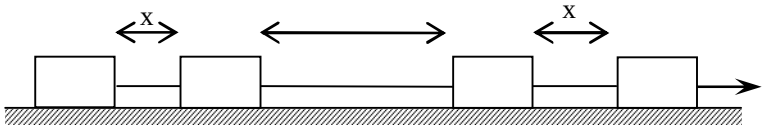
(C) $\frac{1}{2}$ (D) $\frac{4}{3}$

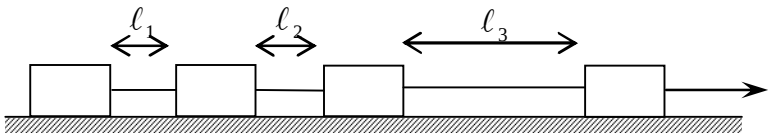
5. A cylinder of height h and radius R is made to move with constant velocity v at small angle θ with its flat bottom or flat top. In a polluted atmosphere having uniform and stagnant dust density ρ. If dust particles stick to cylinder on striking them neglecting any change in size of cylinder, power to be delivered to cylinder to maintain its velocity is

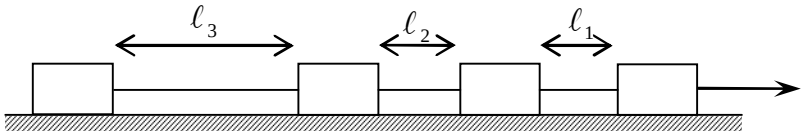
(A) $2\rho v^2 Rh$ (B) $2\rho v^3 Rh$

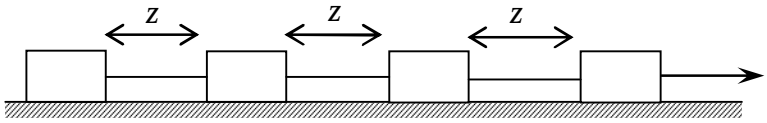
(C) $2\rho v^3 Rh\cos\theta + \rho v^3 \pi R^2 \sin\theta$ (D) None

6. Each of the pictures shows four objects tied together with rubber bands being pulled to right across a horizontal smooth surface by a horizontal force F. All the objects have same mass, all rubber bands are identical naturally and obey Hook's law. Then which of the following possibility of the arrangement is most correct?

(A)


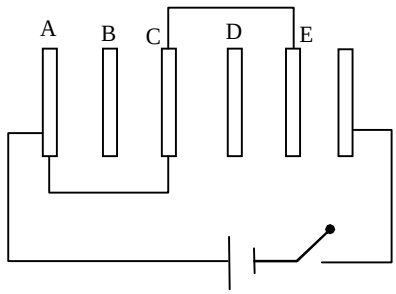
(B)


(C)


(D)


Space for rough work

7. A, B, C, D, E and F are conducting plates each of area A and any two consecutive plates with separation d. The net energy stored in system after the switch S is closed is



(A) $\frac{3\varepsilon_0 AV^2}{2d}$ (B) $\frac{5\varepsilon_0 AV^2}{12d}$

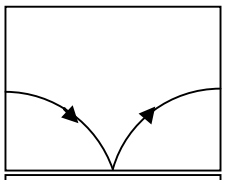
(C) $\frac{\varepsilon_0 AV^2}{2d}$ (D) $\frac{\varepsilon_0 AV^2}{d}$

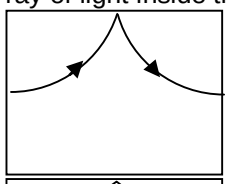
8. When the string of a sonometer wire of length L between the bridges vibrate in the first overtone, the amplitude of vibration is maximum at

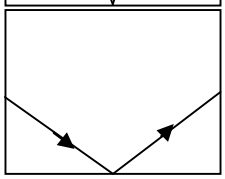
(A) $\frac{L}{2}$ (B) $\frac{L}{4}$ and $\frac{3L}{4}$

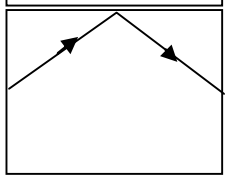
(C) $\frac{L}{6}$, $\frac{3L}{6}$ and $\frac{5L}{6}$ (D) $\frac{L}{8}$, $\frac{3L}{8}$, $\frac{5L}{8}$, $\frac{7L}{8}$

9. A cubical container is filled with a liquid whose refractive index increases linearly from top to bottom. Which of the following represents the path of a ray of light inside the liquid?

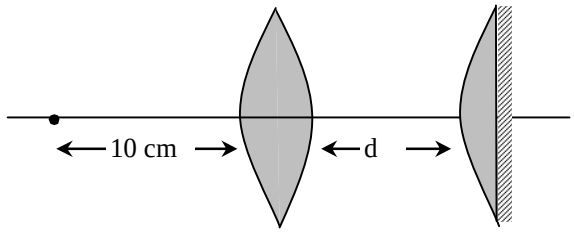
(A) 

(B) 

(C) 

(D) 

10. A convex lens of focal length 20 cm and another plano convex lens of focal length 40 cm are placed coaxially as shown in the diagram. The plano convex lens is silvered at plane surface. If final image of object O is formed on itself, then find the value of d.



(A) 10 cm
(B) 40 cm
(C) 20 cm
(D) 80 cm

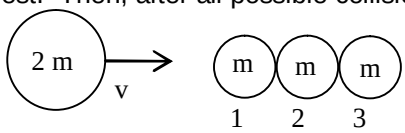
Space for rough work

Multiple Correct Choice Type

This section contains **5 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct.

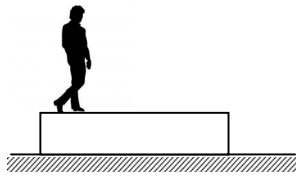
11. A uniformly charged non-conducting ring is rotating with angular velocity ω_1 about its axis in gravity free space and a nearly horizontal but uniform magnetic field B is switched on, then after some time
(A) Angular momentum of the ring must be parallel to its angular velocity
(B) Magnitude of angular momentum may be more than $mR^2\omega_1$
(C) Magnitude of angular momentum must be equal to $mR^2\omega_1$
(D) Angular momentum must be coplanar with angular velocity

12. A steel ball of mass 2m suffers one dimensional elastic collision with a row of three steel balls each of mass m. If ball of mass 2m has velocity v initially and the three balls with small separation were at rest. Then, after all possible collisions



(A) Ball 1 and ball of mass 2m would remain at rest
(B) Ball 2 moves with velocity $\frac{4v_0}{9}$ towards right
(C) Ball 3 moves towards right with velocity $\frac{4v_0}{3}$
(D) Ball of mass 2m moves towards right with velocity one fourth that of ball 1

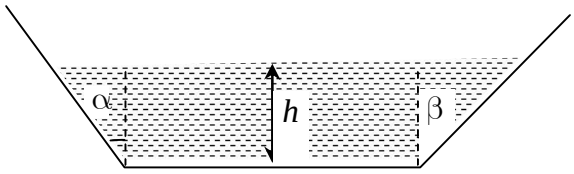
13. A man is standing over a plank which is placed over a smooth horizontal surface. There is sufficient friction between feet of man and plank. Now, man starts running over the plank. Then, work done by friction



(A) On man w.r.t. ground is negative
(B) On man w.r.t. ground is positive
(C) On plank w.r.t. ground is positive
(D) On man w.r.t. plank is zero

Space for rough work

14. A liquid is filled upto height h in a rectangular vessel as shown in figure. Then, find correct option(s):



(A) Only if $\alpha = \beta$ horizontal component of forces on left and right side of inclined faces will be equal and opposite

(B) If $\alpha \neq \beta$ horizontal component of forces on left and right side of inclined faces are not equal

(C) If $\alpha \neq \beta$ still horizontal component of forces on the left and right inclined faces are equal and opposite

(D) If base area is A, then force on bottom by liquid is more than ρghA

15. For an ideal gas at very high temperature,

(A) $C_V = \frac{3R}{2}$ for a monoatomic gas

(B) $C_V > \frac{3R}{2}$ for a monoatomic gas

(C) $C_V < \frac{5R}{2}$ for a diatomic or linear poly atomic gas

(D) $C_V > \frac{5R}{2}$ for a diatomic gas

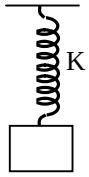
Space for rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. A non-conducting block of mass 1 kg and charge 1 C is connected to a non-conducting spring of force constant $K = \pi^2$ N/m and released. After one second from release a vertical uniform electric field 10 N/C is switched on. If time after switching on the electric field spring returns back to its natural length is T, then find $\frac{1}{T}$.

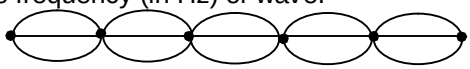


2. A particle of charge q and mass m is projected with velocity 200 m/s at an angle of 60° with x-axis. At the instant speed of projectile becomes minimum a uniform magnetic field $-B_0\hat{k}$ is switched on. If time of flight of particle is T, then find $\frac{2}{T}$. (Given: m = 1 kg, q = 0.1 C, B = 1 T)

3. Figure shows two conducting thin concentric shells of radii r and 3r. The outer shell carries a charge q and the inner shell is neutral. The amount of charge that flows from inner shell to the earth after key K is closed is equal to n^{th} part of charge of outer shell. Then the value of n is ……

4. A vessel contains helium, which expands at constant pressure when 15 KJ of heat is supplied to it. What will be the variation in internal energy of gas in KJ?

5. Two identical sinusoidal waves travel in opposite direction in a 15 m long wire and produce a standing wave in the wire with nodes at ends. If speed of wave is 12 m/s and there are 6 nodes in standing wave, find the frequency (in Hz) of wave.



Space for rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

This section contains **10 multiple choice questions** numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct.

1. Which of the following process is used to remove temporary as well as permanent hardness of water?
   (A) only boiling
   (B) adding lime & boiling
   (C) adding $Ca(OH)_2$
   (D) adding Na_2CO_3

2. When 10 gm mixture of KI and $Fe_2(SO_4)_3$ is treated with 200 cc HCl, gives a solution which consumes 40 ml $\frac{N}{10}$ $Na_2S_2O_3$ solution. Then the correct statement is/are
   (A) Equivalent weight of iodine in the reaction is $\frac{M}{2}$
   (B) Percentage of KI in the mixture is 66.4%
   (C) Sodium thiosulphate is converted to $Na_4S_4O_6$
   (D) Percentage of $Fe_2(SO_4)_3$ is 66.4%

3. Both AgCNS and AgBr are simultaneously dissolved in water. K_{sp}(AgBr) 5×10^{-13}; K_{sp}(AgCNS) = 8.5×10^{-12}. The concentration of Ag^+ in final solution will be
   (A) 4×10^{-7} M
   (B) 2×10^{-7} M
   (C) 3×10^{-6} M
   (D) 6.2×10^{-7} M

4. Compare basic strength of following compounds

Aniline	o-toluidine	m-toluidine	p-toluidine	o-anisidine
1	2	3	4	5

   (A) 1 > 2 > 3 > 4 > 5
   (B) 2 > 5 > 3 > 4 > 1
   (C) 4 > 3 > 1 > 5 > 2
   (D) None of these

Space for rough work

5. $AgNO_3 \xrightarrow{Na_2S_2O_3} \underset{\text{White PPt}}{A} \xrightarrow{H_2O} \underset{\text{Black PPt}}{B}$

Black PPt is

(A) $Na_3\left[Ag(S_2O_3)_2\right]$ (B) $Ag_2S_4O_6$

(C) Ag_2O (D) Ag_2S

6. When glucose is heated with conc. H_2SO_4, charring occurs, it is due to

(A) Reduction of glucose (B) Oxidation of glucose

(C) Dehydration of glucose (D) Mutarotation of glucose

7. IUPAC name of $\left[Fe(O_2)(CN)_4Cl\right]^{4-}$ is

(A) chloridortetracyanodioxoferrate (II) ion

(B) chloridortetracyanoperoxoferrate (II) ion

(C) chloridortetracyanosuperoxoferrate (II) ion

(D) tetracyanochloridosuperoxoferrate (II) ion

8. The $\pi-$acid ligand which uses its d-orbital during synergic bonding in its complex compound is

(A) $\overline{C}N$ (B) PR_3

(C) NO (D) N_2

9. Efficiency of a cell with cell reaction under standard condition is 85%.

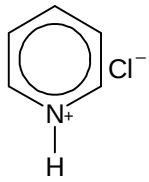
$A_{(s)} + B^{2+}_{(aq)} \rightleftharpoons A^{2+}_{(aq)} + B_{(s)};\ \Delta H^0 = -285kJ$

The E^0_{cell} is:

(A) 1.26 V (B) 1.34 V

(C) 2.96 V (D) 2.07 V

Space for rough work

10. $C_2H_5 - CH_2 - Cl \xrightarrow[Py,\Delta]{DMSO} A + B +$ (has two C atoms per molecule) 

Which is true?

(A) A gives Lucas test

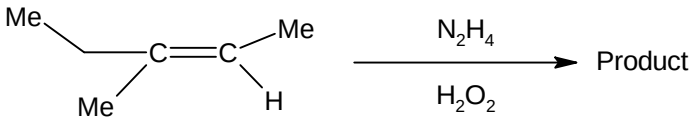
(B) A does not give Schiff's test

(C) B can be used in reductive ozonolysis of alkenes

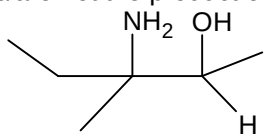
(D) A gives smell of burning S on combustion

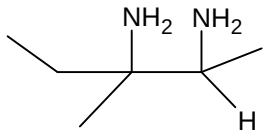
Multiple Correct Choice Type

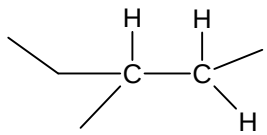
This section contains **5 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct.

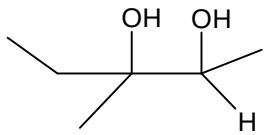
11. 

What is/are not the product of above reaction?

(A) 

(B) 

(C) 

(D) 

12. Which of the following crystal system does/do not have more than one Bravais lattice?

(A) Triclinic (B) Monoclinic

(C) Rhombohedral (D) hexagonal

Space for rough work

13. Which is/are white or not coloured?

(A) $\left[Co(SCN)_4\right]^{2-}$ (B) $\left[Fe(H_2O)_5SCN\right]^{2+}$

(C) $CuSO_4.5H_2O$ (D) anhyd $CuSO_4$

14. Which is/are true for the following reaction?

$$H_5C_2—O—C_2H_5 \xrightarrow[h\nu]{O_2} A + B \xrightarrow{Fe^{2+}} \xrightarrow{KCNS} \text{red color}$$

(A) Compounds A & B are explosives

(B) If compounds A & B are present in ether, then prior to its use, A & B must be removed from ether

(C) Compounds A & B can not be removed by adding Fe^{3+}

(D) When compounds A & B are treated with Fe^{2+} followed by KCNS, no redox reaction is involved

15. $(NaNO_2 + HCl)$ converts an amino acid into hydroxyl acid with retention of configuration. Estimation of N_2 gas evolved in the reaction is the basis of van slyke estimation of amino acids, which of the following can be analysed by this method?

(A) Cysteine (B) Proline

(C) Histidine (D) Valine

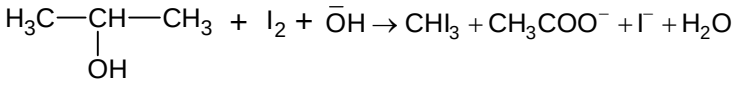
Space for rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. How many moles I_2 is consumed per mole of isopropanol?



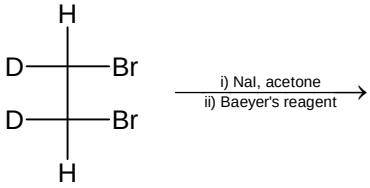
2. How many octahedral voids per unit cell of BCC are present?

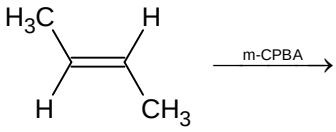
3. The coagulation of 20 ml standard gold sol is completely prevented by addition of 0.04 gm starch solution to it. The gold number of starch solution is 2×10^{x}. Find x.

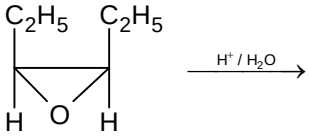
4. If $H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l)$ (Let ΔC_p = 32 J/K and ΔH at 27^oC = –28.58 KJ/mol) reaction is carried out at 87^oC in a fuel cell with 90% efficiency. The electrical energy produced is 3 x a KJ/mol. Find a (to nearest integer). (Let ΔC_p remains same in this temperature range)

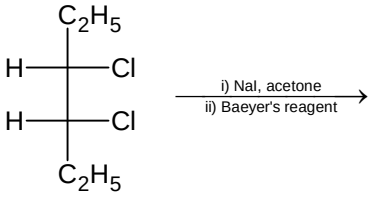
Space for rough work

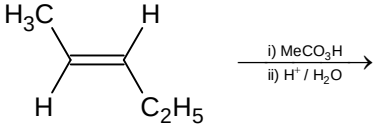
5. In how many of the following reactions end product is meso?

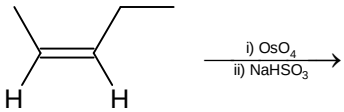
(a) 

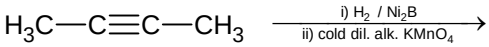
(b) 

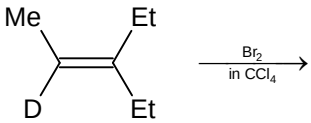
(c) 

(d) 

(e) 

(f) 

(g) 

(h) 

Space for rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **10 multiple choice questions** numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct.

1. If $I=\int_0^{\pi} f(\tan 2x)\cos x\,dx$ and $J=\int_0^{\pi/4} f(\cot 2x)\sin x\,dx$, where $f(-x)=-f(x)$, then $\frac{I}{J}$ is equal to

(A) $\sqrt{2}$ (B) $\frac{1}{\sqrt{2}}$

(C) $2\sqrt{2}$ (D) $\frac{1}{2\sqrt{2}}$

2. If $A=\left[a_{ij}\right]_{2\times 2}$, $B=\left[b_{ij}\right]_{2\times 2}$ such that $a_{ij}=f\left(x^{2i-j-2}\right)$ and $b_{ij}=f\left(x^{2i-j-1}\right)$ and

$$A+B^T=\begin{pmatrix} -x^2 f\left(x+f(1)\right) & \left(1-\frac{1}{x^4}\right) f\left(\frac{1}{x^2}\right) \\ f\left(\frac{1}{x}\right)\left(1-\frac{1}{x^2}\right) & -\frac{1}{x^2} f\left(\frac{1+f(1)}{x}\right) \end{pmatrix}$$, then $\int_x^{1/x} f(t)\,dt$ is always equal to

(A) $f(1)$ (B) $f(1)-1$

(C) $f(1)+1$ (D) None of these

3. If $f(x)=(\cos x)\hat{i}+(\sin x)\hat{j}+(\cos 2x)\hat{k}$, $g(x)=(\tan x)\hat{i}+(\sin 3x)\hat{j}+(\cos 4x)\hat{k}$ and $h(x)=(\cos 3x)\hat{i}+(\sin 5x)\hat{j}+(\cos 6x)\hat{k}$ are three variable vectors given, then $\int_0^{\pi}\left[f(x),\ g(x),\ h(x)\right]dx$, where [.] denotes scalar triple product, is belonging to

(A) $(-1,\ 0)$ (B) $(0,\ 1)$

(C) $(-\infty,\ -1)\cup(1,\ \infty)$ (D) (–1, 1)

Space for rough work

4. If $\vec{a}_1 = a\hat{i} + b\hat{j}$, $\vec{a}_2 = \vec{a}_1 - \left(2\hat{i} + \sqrt{5}\hat{j}\right)$ and $\vec{a}_3 = \vec{a}_1 + \left(2\hat{i} - \sqrt{5}\hat{j}\right)$, then

$$\begin{vmatrix} 1+\vec{a}_1.\vec{a}_1 & 1+\vec{a}_1.\vec{a}_2 & 1+\vec{a}_1.\vec{a}_3 \\ 1+\vec{a}_2.\vec{a}_2 & 1+\vec{a}_2.\vec{a}_2 & 1+\vec{a}_2.\vec{a}_3 \\ 1+\vec{a}_3.\vec{a}_1 & 1+\vec{a}_3.\vec{a}_2 & 1+\vec{a}_3.\vec{a}_3 \end{vmatrix}$$ is equal to

(A) 0
(B) 20
(C) 40
(D) 80

5. $\int_0^{\pi/2} \frac{\sin^3 x\, dx}{(1+\tan x)} =$

(A) $\frac{1}{2} - \frac{1}{2\sqrt{2}} \ln\left(\sqrt{2}+1\right)$
(B) $\frac{1}{2} + \frac{1}{2\sqrt{2}} \ln\left(\sqrt{2}+1\right)$
(C) $\frac{1}{2} - \frac{1}{\sqrt{2}} \ln\left(\sqrt{2}+1\right)$
(D) None of these

6. If $|f(x)| \le |\sin \pi x|$ and $g(x) = f'(x)$, then maximum value of $|g(n)|$ for $n \in I$ is

(A) 0
(B) 1
(C) π
(D) 2π

7. If $f(x) = \tan x + \tan^2 x \tan 2x$ and $g(n) = \sum_{n=0}^{n=n} f\left(2^n\right)$, then $g(2013) - \tan\left(2^{2014}\right)$ is equal to

(A) 0
(B) tan1
(C) $-$tan1
(D) None of these

Space for rough work

8. The value of product

$$\frac{\sin\frac{\theta}{2}}{\tan^2\frac{\theta}{2}\tan\theta+\tan\frac{\theta}{2}}\times\frac{\sin\frac{\theta}{2^3}}{\tan^2\frac{\theta}{2^3}\tan\frac{\theta}{2^2}+\tan\frac{\theta}{2^3}}\times.....\frac{\sin\frac{\theta}{2^n}}{\tan^2\frac{\theta}{2^n}\tan\frac{\theta}{2^{n-1}}+\tan\frac{\theta}{2^n}}\times..... \text{ upto } \infty \text{ is}$$

(A) 1 for $\theta=\frac{\pi}{2}$ (B) 1 for $\theta=\frac{\pi}{4}$

(C) $\frac{1}{2\pi}$ for $\theta=\frac{\pi}{4}$ (D) $\frac{2}{\pi}$ for $\theta=\frac{\pi}{4}$

9. If $f(x)=\frac{\alpha x+1}{x-\beta}\forall x\in R-\{\beta\}$ & $\alpha\beta\neq1$ & $\alpha\neq1$ is a self – inverse function such that $\frac{f(4)}{4}=\frac{f(12)}{12}=f\left(\frac{1+\beta}{1-\alpha}\right)$, then "$\beta$" is equal to

(A) 2 (B) 4
(C) 8 (D) 16

10. If $f(x)=x+1$ & $g^{-1}(x)=x^3+x+1$, let gof(x) = h(x), then $\lim\limits_{x\to1}\frac{h^{-1}(x)-h^{-1}(1)}{x-1}$

(A) 2 (B) 4
(C) 1 (D) 8

Multiple Correct Choice Type

This section contains **5 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct.

11. If $\int_1^2\frac{2x^2-x^{-1}+\sqrt{x}}{\left(1+x^3\right)}dx=\ln\left(\frac{p}{q}\right)+\frac{2}{3}\cos^{-1}\left(\frac{1}{m}\right)-\frac{\pi}{6}$, $p,q\in I^+$, HCF (p, q) = 1, then

(A) $p=m^2-1$ (B) $p=m^2$

(C) $q=m+1$ (D) $pq=4m^2$

Space for rough work

12. If a singular matrix $A=\left[a_{ij}\right]_{2\times 2}$ always commute with $B=\begin{pmatrix}1 & 1\\ 2 & 1\end{pmatrix}$, then

(A) $a_{11}=a_{22}$ (B) $a_{11}=\sqrt{2}a_{12}$

(C) $a_{21}=2a_{12}$ (D) $a_{11}=\sqrt{2}a_{21}$

13. If $\log_k a=a$ and $\log_k b=b$ for exactly two distinct positive real numbers a and b, then k can't be equal to

(A) $e^{\frac{1}{2e}}$ (B) $e^{\frac{2}{e}}$

(C) $e^{\frac{1}{2}}$ (D) $e^{\frac{1}{3}}$

14. If $\Delta(x)=\begin{vmatrix}\cos x+1 & \cos^2 x & 1+\sin^2 x\\ -\sin x & -\sin 2x & \sin 2x\\ \sin x & 2^2\sin 2x & -2^2\cos 2x\end{vmatrix}$, then

(A) $\int_0^{\pi/2}\Delta(x)dx=4$ (B) $\int_0^{\pi/2}\Delta(x)dx=-4$

(C) $\int_{\pi/2}^{\pi}\Delta(x)dx=4$ (D) $\int_{\pi/2}^{\pi}\Delta(x)dx=-4$

15. If curve $A\equiv\{z:|z-3|+|z+3|=8\}$ & $B\equiv\{z:|z-3|=K\}$, $K\in R^+$ touches the curve 'A' internally, and given another curve $C\equiv\{z\,||z-3|-|z+3||=4\}$, then;

(A) A & C intersect orthogonally & k = 1
(B) K = 1
(C) K = 3/2
(D) B touches also C internally

Space for rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. If $I=\int_1^2 \frac{x^3-3x^2}{\ln(x-1)}dx$ and $J=\int_2^3 \frac{(3x-5)}{\ln(x-2)}dx$, then e^{I+J} equals …….

2. Position vector of a point A is $\left(a\hat{i}+b\hat{j}+c\hat{k}\right)$ and $a^2+b^2+c^2=4$; and position vector of another point B is $\left(p\hat{i}+q\hat{j}+r\hat{k}\right)$ and $p^2+q^2+r^2=9$. Given $\angle AOB=2\cos^{-1}\frac{1}{5}$, where O is origin. If AQ is perpendicular to angle bisector of $\angle AOB$ i.e. line OQD where D lies on AB, then value of $25|QD|$ = ?

3. If $\frac{3x^2+1}{\sqrt{x^4+x^2}}\geq f(x)$, where $f(x)$ is a constant function in $0<x\leq\sqrt{2}$, then minimum value of $\int_0^{\sqrt{2}} \frac{(3x^2+1)dx}{\sqrt{x^4+x^2}}$ is ……

4. If $\left|\text{adj}\left(\text{adj}\left(\text{adj } A\right)\right)\right|=\left|\text{adj } (A)\right|^4$, then number of entries in diagonal of *A* is ……..

5. If $AB=BA=I$ such that $A=\left[a_{ij}\right]_{2\times2}$, $B=\left[b_{ij}\right]_{2\times2}$, $a_{ii}=b_{ii}$, $a_{12}=-b_{12}$, $a_{21}=-b_{21}$ and a_{ij} are co prime natural numbers, then $|a_{21}-a_{12}|=$ …….

Space for rough work